

NITIN KUMAR PATEL

IIIT Bhopal, Madhya Pradesh

+91 8103497022 | nitinpatel6875@gmail.com | GitHub | LinkedIn

Profile Summary

Aspiring **AI Engineer** with hands-on experience building **LLM-powered applications** using **Retrieval-Augmented Generation (RAG)**, **Hybrid Retrieval (Dense + Sparse + Re-ranking)**, **LangChain**, and **LangGraph**. Skilled in designing **production-style backend systems** with **FastAPI**, **vector databases**, **JWT authentication**, and **SQL databases**, with additional experience in classical **Machine Learning** and **Deep Learning** for predictive analytics and computer vision.

Education

Indian Institute of Information Technology, Bhopal

2025 – 2029

B.Tech in Computer Science and Engineering

Experience

AI and Data Analytics Intern — XyLofy AI

2026

- Trained and evaluated machine learning models for predictive analytics and fraud detection on structured datasets.
- Developed a **Transaction Fraud Detection** system using **XGBoost**; resolved class imbalance with **SMOTE** and optimized model performance using **ROC-AUC** analysis.

[View Demo](#)

Projects

HIA: Health Insights Agent — Python, FastAPI, LangChain, LangGraph

2026 | [View Demo](#)

- Built a full-stack conversational AI application that lets users upload medical reports (PDF) and understand them through natural, context-aware conversations.
- Designed a **Hybrid RAG pipeline** combining **dense retrieval (ChromaDB)**, **sparse retrieval (BM25)**, and **FlashRank re-ranking** to improve retrieval quality and response accuracy.
- Implemented **JWT-based authentication** and a **PostgreSQL (Neon) + SQLAlchemy** data layer for secure, user-isolated report storage, and deployed the LangGraph-powered backend and frontend on **Render**.

C-RAG: Conversational RAG for PDFs — Python, LangChain, LLMs

2026 | [View Demo](#)

- Built **C-RAG (Conversational Retrieval-Augmented Generation)**, an LLM-powered system enabling contextual question answering over PDF documents using semantic retrieval.
- Implemented **LangChain**, vector embeddings, chunking, and retrieval pipelines to generate accurate responses from uploaded PDFs.

Brain Tumor Detection — TensorFlow, ResNet50V2

2026 | [View Demo](#)

- Developed a multi-class medical image classification model using **ResNet50V2**, achieving **91% validation accuracy**.
- Built and deployed an end-to-end AI inference pipeline using **TensorFlow** and **Streamlit**.

Technical Skills

Programming: Python, C++

LLM & GenAI: RAG, Hybrid RAG (Dense + Sparse + Re-ranking), LangChain, LangGraph, Prompt Engineering, AI Agents, Embeddings

Retrieval & Vector DBs: ChromaDB, FAISS, BM25, FlashRank

Backend & APIs: FastAPI, REST API Development, JWT Authentication, SQLAlchemy, PostgreSQL (Neon)

AI/ML: Machine Learning, Deep Learning, Natural Language Processing (NLP), CNNs, Transfer Learning, TensorFlow, Keras, Scikit-learn, XGBoost

LLM Providers: Groq LLM, Google Gemini, Hugging Face Embeddings

Data Tools: NumPy, Pandas, Excel, Power BI

Tools & Deployment: Git, GitHub, Cloud Platforms (Render), Model Deployment, Streamlit, Jupyter Notebook, VS Code

Concepts: LLM Fine-Tuning, Semantic Search, Information Retrieval, MLOps, SMOTE, ROC-AUC

Leadership

First Year Representative — Spark Club, IIIT Bhopal

2025 – Present

Achievements

- Winner — AWS Track (Hackathon)
- 2★ Rating on CodeChef
- Solved 100+ DSA Problems on LeetCode